Kyu Hyun Lee

kyuhyun.lee@ucsf.edu | Google Scholar | khl02007.github.io/blog

Academic Appointments

Postdoctoral Scholar, University of California, San Francisco 2020–present
Advisor: Loren M. Frank

Education

California Institute of Technology 2013–2020
Ph.D., Biochemistry and Molecular Biophysics option
Advisor: Markus Meister

Pomona College 2007–2011
B.A., Chemistry (minor: Mathematics), *magna cum laude*
Honors: Phi Beta Kappa; Robert Rowan III Memorial Award

Research Funding and Fellowships

NIH/NEI Pathway to Independence Award (K99/R00) 2025–present
Principal Investigator • K99EY036953

Helen Hay Whitney Foundation Postdoctoral Fellowship 2022–2025

Tianqiao and Chrissy Chen Institute Graduate Innovator Grant 2018

Caltech-Amgen Graduate Fellowship 2014–2015

Publications and Scholarly Products

Preprints

Lee, K., Adenekan, P., Gao, F., Zhou, J., Hernandez, J., Yorita, A., Haque, R., Kay, K., Scanziani, M., and Frank, L.M. (2026). A cognitive representation in primary visual cortex modulated by vision. *bioRxiv*. doi: 10.64898/2026.08.29.748029.

Book

Meister, M., **Lee, K.**, and Portugues, R. (2025). *Mathematics in Biology*. MIT Press.

Peer-reviewed publications

1. Guidera, J.A., Gramling, D.P., Comrie, A.E., Joshi, A., Tseng, S.Y., Denovellis, E.L., Smyth, C.N., **Lee, K.**, Zhou, J., Thompson, P., Hernandez, J., Yorita, A., Haque, R., Kirst, C., and Frank, L.M. (2026). Regional specialization in prefrontal cortex manifests in the reliability of task progression codes. *Neuron* 114(6), 1152–1161.e8.
2. **Lee, K.**[*], Denovellis, E.[*], Ly, R., Magland, J., Soules, J., Comrie, A.E., Gramling, D.P., Guidera, J.A., Nevers, R., Adenekan, P., Brozdowski, C., Bray, S., Monroe, E., Bak, J., Coulter, M., Sun, X., Broyles, E., Shin, D., Chiang, S., Holobetz, C., Tritt, A., Rübel, O., Nguyen, T., Yatsenko, D., Chu, J., Kemere, C., Garcia, S., Buccino, A., and Frank, L.M. (2025). Spyglass: a framework for reproducible and shareable neuroscience research. *eLife* 14, RP108089.
3. **Lee, K.**[*], Ni, Y.[*], Colonell, J., Karsh, B., Putzeys, J., Pachitariu, M., Harris, T.D., and Meister, M. (2021). Electrode pooling can boost the yield of extracellular recordings with switchable silicon probes. *Nature Communications* 12, 5245.
4. **Lee, K.**, Tran, A., Turan, Z., and Meister, M. (2020). The sifting of visual information in the superior colliculus. *eLife* 9, e50678.

5. **Lee, K.**, Humbarger, S., Bahnvadia, R., Sazinsky, M.H., and Crane, E.J., III (2014). Characterization of the mechanism of the NADH-dependent polysulfide reductase (Npsr) from *Shewanella loihica* PV-4: Formation of a productive NADH-enzyme complex and its role in the general mechanism of NADH and FAD-dependent enzymes. *Biochimica et Biophysica Acta - Proteins and Proteomics* 1844(9), 1708–1717.

6. Warner, M.D., Lukose, V., **Lee, K.**, Lopez, K., Sazinsky, M.H., and Crane, E.J., III (2011). Characterization of an NADH-dependent persulfide reductase from *Shewanella loihica* PV-4: Implications for the mechanism of sulfur respiration via FAD-dependent enzymes. *Biochemistry* 50(2), 194–206.

Peer-reviewed conference proceeding

Hermiz, J., Joseph, E., **Lee, K.**, Baldacci, I.A., Chung, J.E., Frank, L.M., Bouchard, K.E., and Denes, P. (2021). The impact of reducing signal acquisition specifications on neuronal spike sorting. *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 5914–5918.

Patent

Meister, M., **Lee, K.**, and Ni, Y. (2024). Methods and systems for electrode pooling. U.S. Patent No. 12,097,033.

*Equal contribution.

Invited Talks and Oral Presentations

Navigating Educational Trajectories in Neuroscience (NIH R25 Program) 2024, 2025
University of Oregon

Dynamic Brain Circuits Cluster Databinge 2024
University of British Columbia, Djavad Mowafaghian Centre for Brain Health

"Reproducible Workflows for Collaborative Neuroscience" symposium 2023
BRAIN Initiative Meeting, NIH

Spike Sorting Workshop 2018
Janelia Research Campus

Junior Scientist Workshop on Neural Circuits and Behavior 2017
Janelia Research Campus

Selected Poster Presentations

"Visual cues shape an internal task-state representation in primary visual cortex" 2026
Lee, K., Adenekan, P., Gao, F., Zhou, J., Hernandez, J., Yorita, A., Haque, R., Kay, K., Scanziani, M., and Frank, L.M.
Neurobiology of Cognition Gordon Research Conference, Waterville Valley, New Hampshire

"A high-resolution spatial map in the primary visual cortex of freely behaving rats" 2025
Lee, K., Adenekan, P., Gao, F., Yorita, A., Haque, R., Eden, U., Scanziani, M., and Frank, L.M.
Neuroscience 2025, Society for Neuroscience, San Diego, California

"Cognitive representations in primary visual cortex" 2025
Lee, K., Adenekan, P., Gao, F., Yorita, A., Haque, R., Eden, U., Scanziani, M., and Frank, L.M.
Neural Coding and Dynamics, Lake Conferences, Seattle, Washington

"Spyglass: a data analysis framework for reproducible and shareable neuroscience research" 2022
Lee, K., Denovellis, E., Ly, R., Magland, J., Soules, J., Comrie, A., Guidera, J., Nevers, R., Gramling, D., Adenekan, P., Coulter, M., Bak, J., Monroe, E., Tritt, A., Rübel, O., Nguyen, T., Yatsenko, D., Chu, J., Kemere, C., Garcia, S., Buccino, A., Aery Jones, E., Giocomo, L., and Frank, L.
Neuroscience 2022, Society for Neuroscience, San Diego, California

"Visual sifting in the superior colliculus" 2019
Lee, K., Tran, A., Turan, Z., and Meister, M.
Simons Collaboration on the Global Brain, New York City, New York

Workshops Attended

NWB Developer Days 2023
Janelia Research Campus

NeuroData ReHack 2022
Allen Institute for Brain Science

Spike Sorting Hackathon 2022
Center for Computational Biology, Flatiron Institute

Mountainsort User Workshop 2018
Center for Computational Biology, Flatiron Institute

Research Mentoring

Denisse Morales-Rodriguez, Graduate Student, UCSF 2024–present
Optogenetic manipulation of mPFC during sleep.

Scott Gao, Research Associate, UCSF 2024–present
Hippocampus–visual cortex interactions.

Philip Adenekan, Research Associate, UCSF 2022–2024
Hippocampus–visual cortex interactions.

Yangsun Hwang, Rotation Student, UCSF 2023
Chronic tracking of neurons recorded with polymer probes.

Roshanak Gonzalez, Rotation Student, UCSF 2022
Testing an optoelectronic neural probe.

Daryl Lawrence, Rotation Student, UCSF 2021
Denoising signals from a neural-recording amplifier chip.

Teaching Experience

Teaching Assistant
Division of Biology and Biological Engineering, California Institute of Technology

Principles of Neuroscience (Fall term) 2018

Tools of Neurobiology (Fall term) 2016

Mathematics in Biology (Fall term) 2015

Brain Circuits (Winter term) 2014

References

Markus Meister meister@caltech.edu
Anne P. and Benjamin F. Biaggini Professor of Biological Sciences, Caltech

Loren M. Frank loren.frank@ucsf.edu
HHMI Investigator, Professor of Physiology, University of California, San Francisco

Massimo Scanziani massimo@ucsf.edu
HHMI Investigator, Professor of Physiology, University of California, San Francisco